



# Experimental and Cohort Studies

APHS 20203 - Epidemiology & Biostatistics

# Review: Association vs Causality

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What is an **association**?

- The distribution of an outcome differs across levels of an exposure.
  - For example, the outcome may be more common in one group than another.
- Knowing something about **X** helps us predict something about **Y**.

# Review: Association vs Causality

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What distinguishes an **association** from a **causal relationship**?

- **Counterfactual definition:** The outcome would differ for the same person if their exposure had been different.
- **Intervention definition:** Intervening to change the exposure would change the outcome or its probability. Said another way, X is a cause of Y if we can change Y by manipulating X.
- **Helpful question:** If I changed X and held everything else constant, would I expect Y to change?

# Review: Deterministic vs Probabilistic Causality

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What distinguishes **deterministic** causality from **probabilistic** causality?

- **Deterministic:** Exposure **always** produces the outcome.
- **Probabilistic:** Exposure **increases the likelihood** of the outcome.

# Two Categories of Analytic Studies

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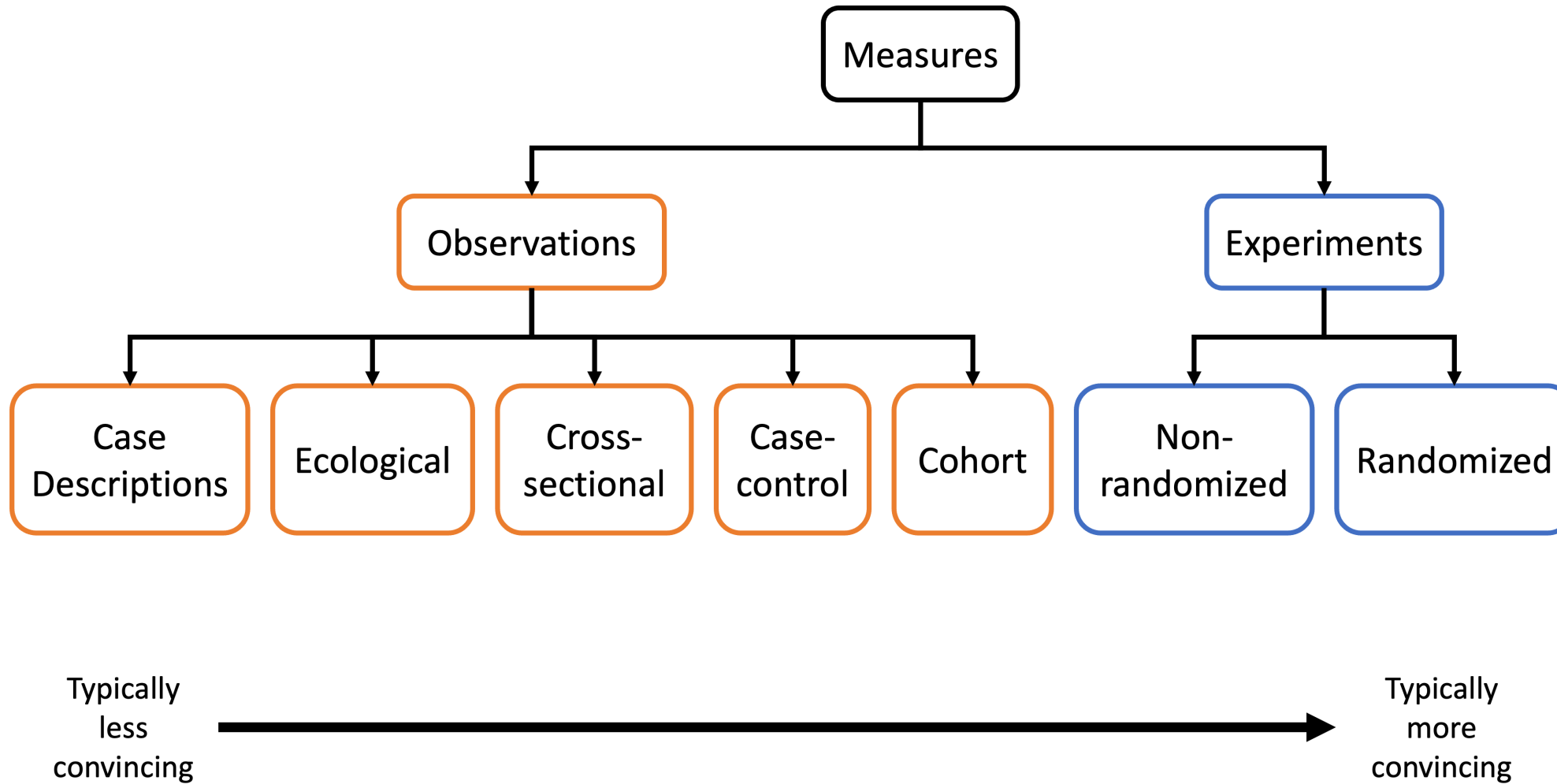
## Observational design

- Investigator **does not assign exposure**
- Observes existing differences in exposure and outcome
- Examples include cross-sectional, cohort, and case-control studies

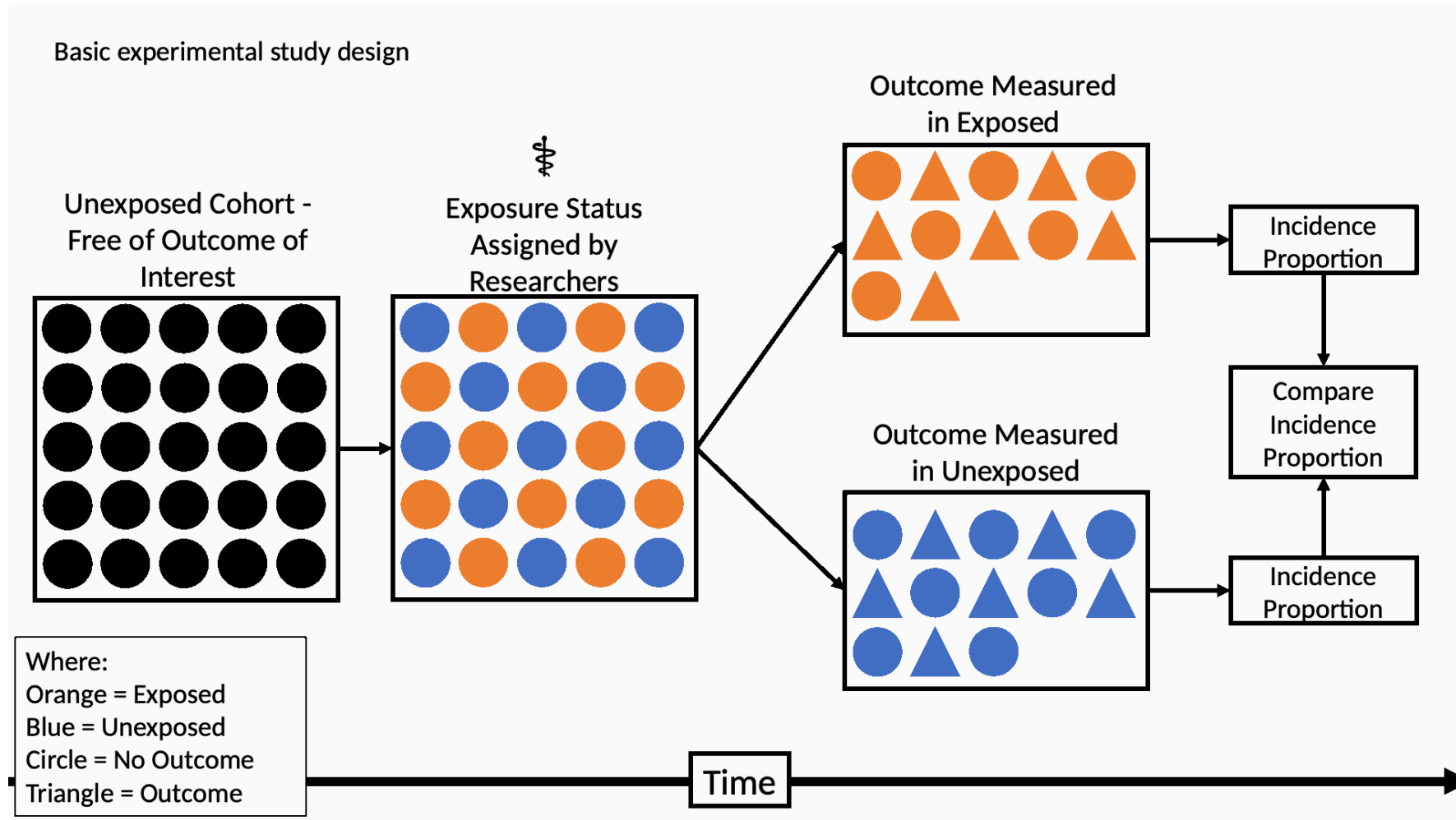
## Experimental design

- Investigator **assigns exposure or intervention**
- Often uses **randomization**
- Examples include randomized controlled trials (RCTs)

# Hierarchy of Study Designs



# Basic Experimental Study Design



# RCT: Randomization

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**Randomization** assigns participants to treatment groups using a chance process.

## Why randomization matters

- Breaks the link between **baseline causes of the outcome** and **treatment assignment**.
- Protects against confounding by **measured and unmeasured factors**, without needing to identify them all.
- Produces **exchangeable groups on average**: groups expected to have the same outcome risk if given the same treatment.

## Take-home message

Randomization protects against **confounding on average**. Chance differences between groups can still occur in a particular trial.

# Practice: Baseline Group Comparability

Participants were randomly assigned to aspirin (**8,517**) or placebo (**8,514**).

| Baseline characteristic   | Aspirin, n | Placebo, n | Ratio* |
|---------------------------|------------|------------|--------|
| Past cigarette smoking    | 3,321      | 3,320      | 1.00   |
| Current cigarette smoking | 847        | 841        | 1.01   |
| Diabetes mellitus         | 196        | 195        | 1.01   |

*Ratio of the proportion with the characteristic in the aspirin group to the proportion in the placebo group; labeled RR in the source table.*

1. What does a ratio close to **1** tell us about a baseline characteristic?
2. Do these rows describe treatment outcomes or participants **before treatment**?
3. Do similar proportions for these three characteristics establish that the groups are identical in every way?

# Interpreting the Baseline Table

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1. Ratios close to **1** indicate similar proportions with these **measured baseline characteristics**.
2. These rows contain baseline comparisons, so they describe participants **before treatment**.
3. Similarity in these measured characteristics does **not** establish similarity in every measured or unmeasured characteristic.
  - Randomization supports comparability **on average**; chance differences can remain in a particular trial.

# Internal and External Validity

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- **Internal validity:** Does the study provide an unbiased estimate of the association *among the people studied*?
  - Comparable groups at baseline, like the aspirin and placebo groups we just examined, support internal validity.
- **External validity:** Do the findings apply to people *outside the study*?
  - This depends on who enrolled. Would the aspirin results hold for people who differ from the trial participants in age, sex, or health?

# RCTs: Strengths & Limitations

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## Strengths

- Randomization improves **group comparability**
- Strong **internal validity**
- Typically provides strongest evidence for the **causal effects** of interventions

## Limitations

- Often **expensive** and time-consuming
- May have **limited generalizability** to real-world populations
- Not always **ethical or feasible**

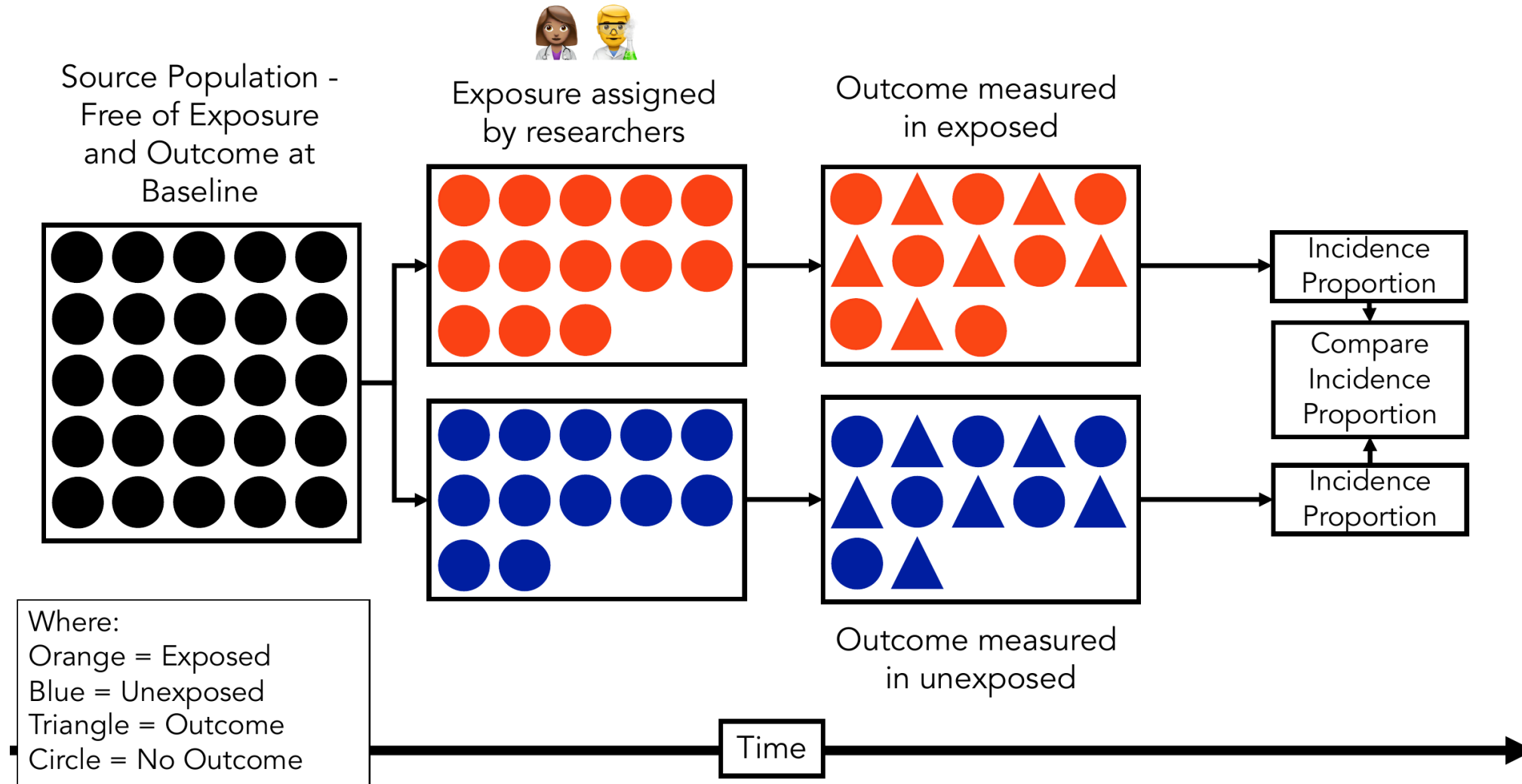
# What Is a Cohort?

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A **cohort** is an identified group of people whose membership is defined by a shared characteristic or event.

- **Example:** Everyone who graduated from TCU in 2026.
- We can identify **who belongs to the group**.
- In epidemiology, we often follow a cohort over time to observe health outcomes.
- **Key difference between a population and a cohort:** Every member of the cohort is enumerated (i.e., known or listed). That is not necessarily true of a population.

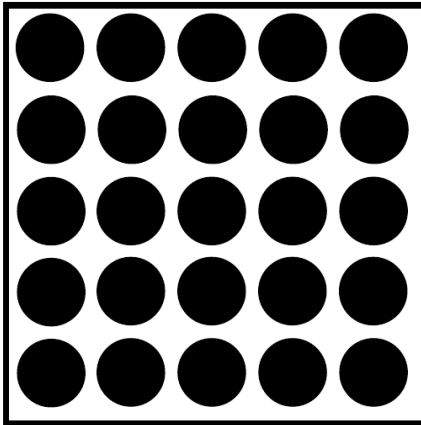
# Basic Experimental Study Design



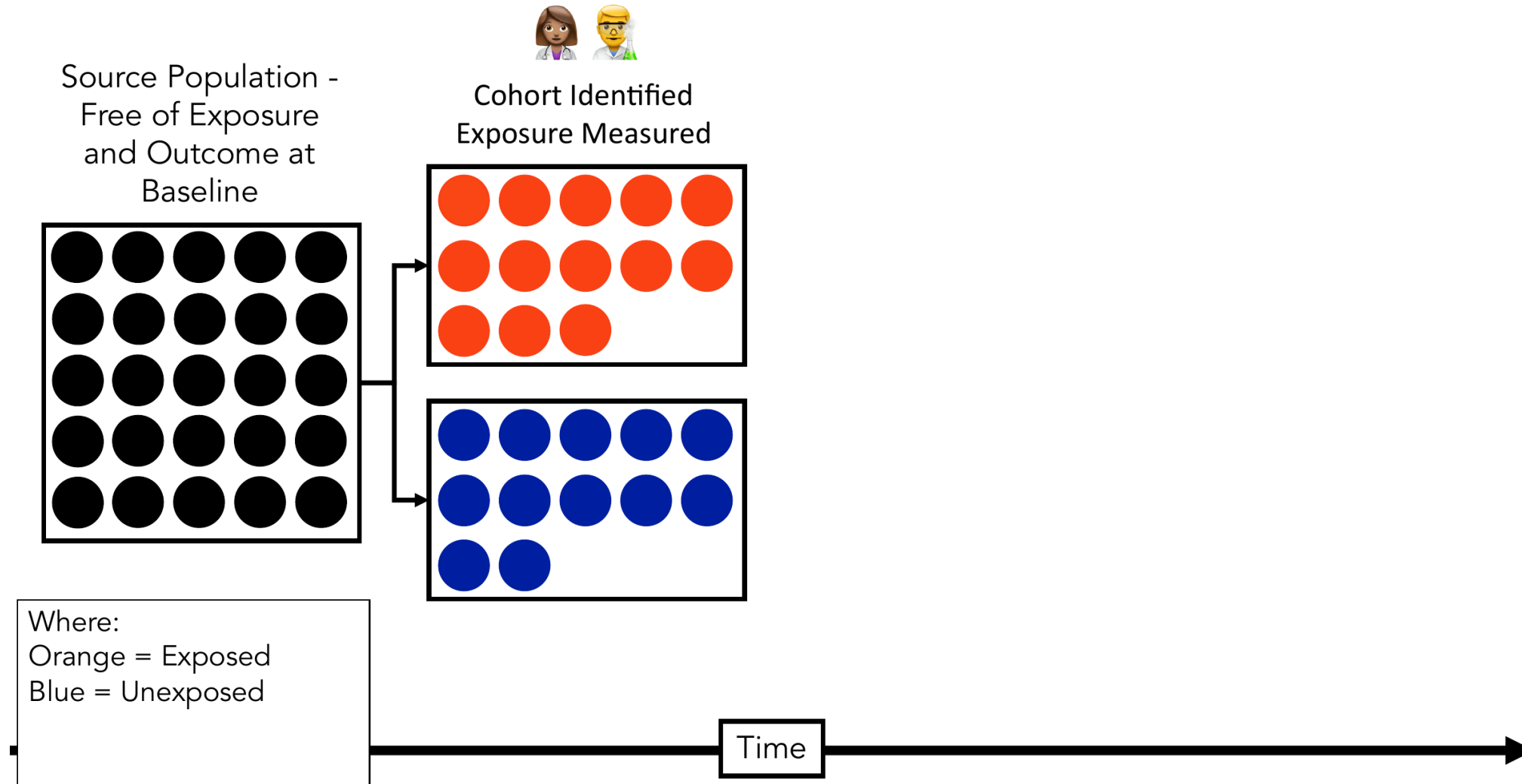
# Common Prospective Cohort Study Design

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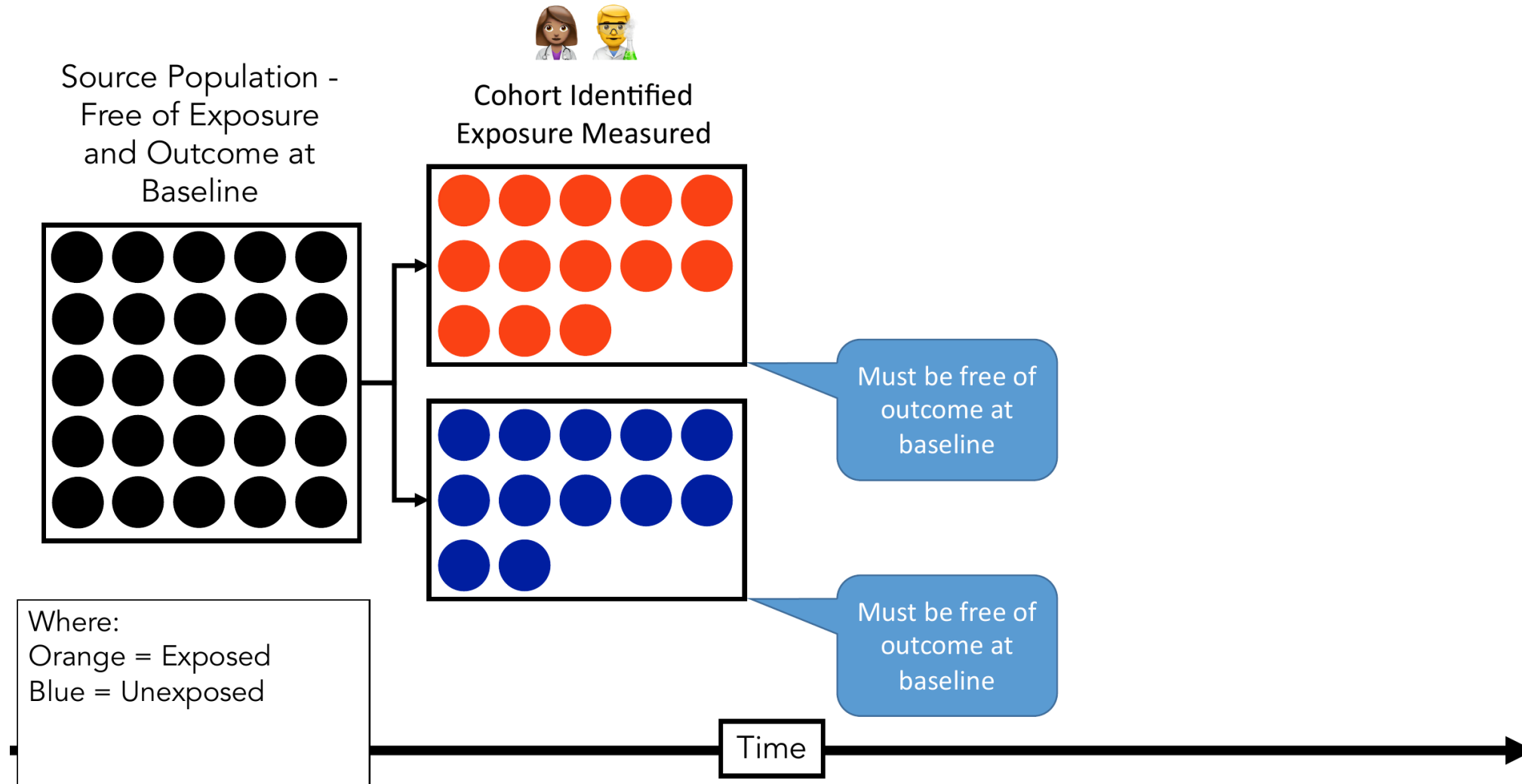
Source Population -  
Free of Exposure  
and Outcome at  
Baseline



# Common Prospective Cohort Study Design



# Common Prospective Cohort Study Design

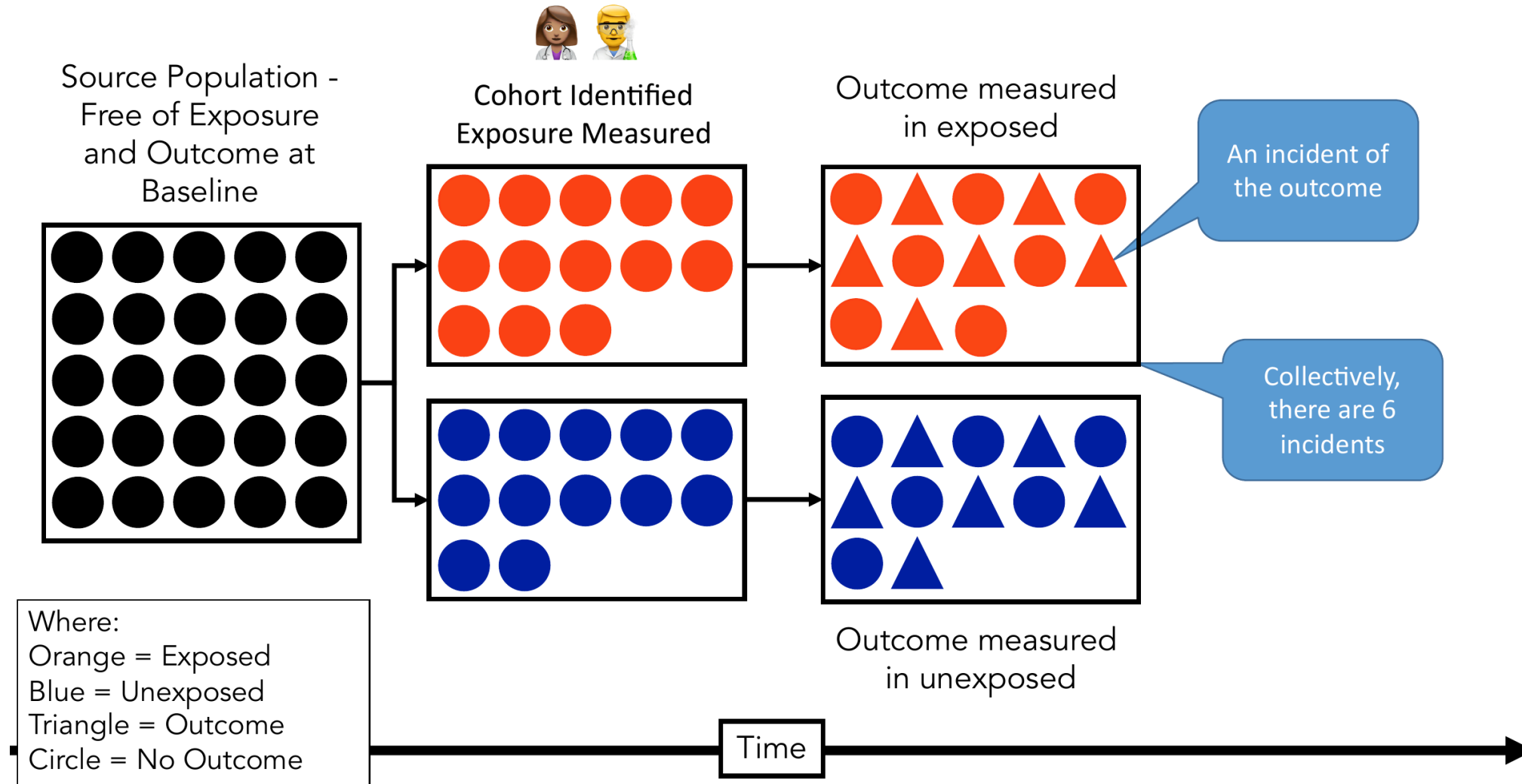


# Population at Risk

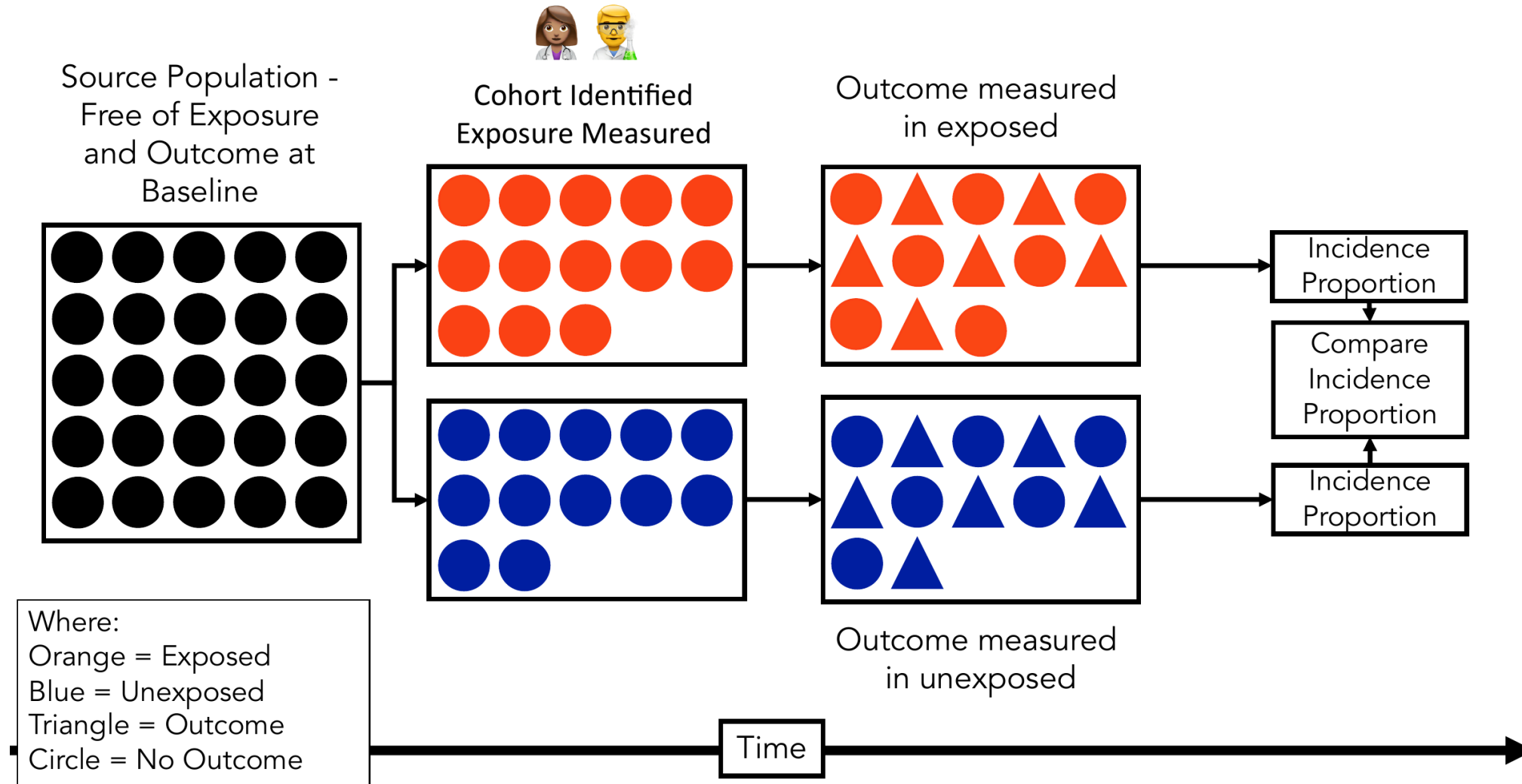
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- All members of the cohort should be **at risk of developing** the outcome of interest.
- Implies that all members should be **free of the outcome** of interest at the outset of follow-up.

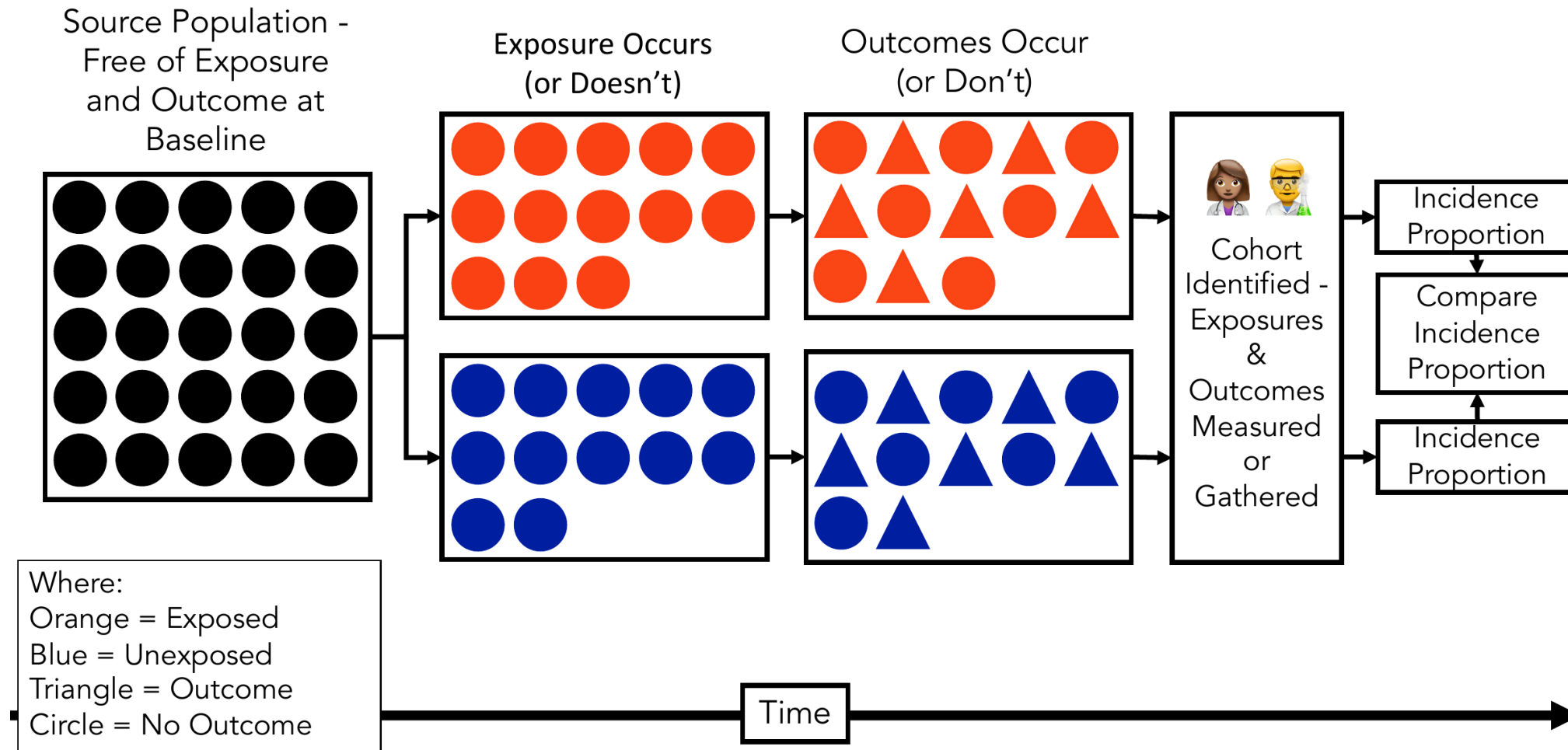
# Common Prospective Cohort Study Design



# Common Prospective Cohort Study Design



# Common Retrospective Cohort Study Design



# Prospective vs. Retrospective Cohorts

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- Both follow the same direction: **Exposure → Outcome**
- Both sample on the **Exposure**
- **Prospective Cohort**
  - **Exposure measured in the present**
  - Outcomes occur in the future
- **Retrospective Cohort**
  - **Exposure and outcome already occurred**
  - Uses existing records
  - Limited by data quality and missing variables
- **Key Difference:** Timing of data collection — not sampling design.

# When to Conduct a Cohort Study

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- **Exposure is rare:** efficient for studying uncommon exposures
- **Incidence must be measured:** can directly calculate risk and RR
- **Temporal sequence is important:** exposure clearly precedes outcome
- **Multiple outcomes are of interest:** one exposure → several possible diseases
- **Follow-up is feasible:** resources, time, and stable population available
- **Design Logic:** Cohort studies are strongest when you need clear **exposure → outcome** tracking over time.

# The Framingham Heart Study

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- **Study Design:** Prospective **Cohort Study**
- **What Was Done:**
  - Enrolled **5,209 adults** to study the development of **cardiovascular disease (CVD)**
  - Measured baseline exposures (e.g., blood pressure, cholesterol, smoking)
  - Followed participants over time for **incident CVD**
- **Why This Design Was Appropriate:**
  - Needed to measure **incidence**
  - Required clear **exposure → outcome** sequence
  - Allowed one exposure to be linked to **multiple outcomes**
  - Long-term follow-up was essential
- **Key Lesson:** Cohort studies are powerful for identifying **risk factors over time.**

# Potential Biases in Cohort Studies

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- **Selection Bias**
  - **Loss to follow-up (attrition):** selective dropout can cause **attrition bias**
  - *Example:* Participants who developed severe illness were less likely to return for follow-up visits → risk may be underestimated.
- **Information Bias**
  - Misclassification of **exposure**
  - *Example:* Smoking status or diet self-reported inaccurately at baseline.
- **Confounding**
  - A third variable related to both exposure and outcome
  - *Example:* Age is associated with higher blood pressure and higher CVD risk.
- **Key Question:** Which bias is most important in a long prospective cohort?

# Assessing Risk in Cohort Studies

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- **Risk (Incidence Proportion):** Number of new cases  $\div$  population at risk
- **Relative Risk (RR):** Risk in exposed  $\div$  Risk in unexposed
- **Absolute Risk Difference (ARD):** Risk in exposed – Risk in unexposed
- **Key Distinction:**
  - RR shows strength of association
  - ARD shows public health impact

# A Prospective Cohort Study Example

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- **Research Question:** Is passive tobacco smoke exposure associated with increased respiratory infections in children?
- **Study Design:**
  - Enroll **disease-free children (<12 years)**
  - Classify by **household smoking exposure**
    - Exposed vs Not exposed
- **Follow forward in time**
  - Measure **incident infections**
- **Potential Confounders:**
  - Air pollution
  - Socioeconomic status
  - Crowded housing
- **Next Step:** Calculate **Risk, RR, and ARD**

# Reading a 2×2 Table

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|             | Infection (+) | Infection (–) | Total |
|-------------|---------------|---------------|-------|
| Exposed     | A             | B             | A + B |
| Not Exposed | C             | D             | C + D |

- **Rows** identify exposure status; **columns** identify infection status.
- Each letter represents a **count of people**. For example, **A** is the number of exposed children who developed an infection.
- **A + B** is the total exposed group; **C + D** is the total unexposed group.
- These are the denominators for calculating risk in each group.

## Exercise: Risk (Incidence Proportion)

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|             | Infection (+) | Infection (–) | Total |
|-------------|---------------|---------------|-------|
| Exposed     | 100           | 400           | 500   |
| Not Exposed | 30            | 470           | 500   |

### Calculate Risk (Incidence Proportion)

- Risk in the exposed =  $A / (A + B)$
- Risk in the unexposed =  $C / (C + D)$

# Interpretation: Risk (Incidence Proportion)

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- **Risk in the exposed =  $100 / 500 = 0.20 = 20\%$** 
  - 20 out of every 100 exposed children developed an infection during follow-up.
- **Risk in the unexposed =  $30 / 500 = 0.06 = 6\%$** 
  - 6 out of every 100 unexposed children developed an infection during follow-up.

## Exercise: Relative Risk (RR)

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|             | Infection (+) | Infection (–) | Total |
|-------------|---------------|---------------|-------|
| Exposed     | 100           | 400           | 500   |
| Not Exposed | 30            | 470           | 500   |

### Calculate Relative Risk (RR)

- $RR = \text{Risk in the exposed} \div \text{Risk in the unexposed}$

## Interpretation: Relative Risk (RR)

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- $RR = 0.20 \div 0.06 \approx 3.33$
- Exposed children had **about 3.33 times the risk** of infection compared with unexposed children during follow-up.
- Passive smoke exposure was associated with a **substantial increase in risk** during follow-up.

## Exercise: Absolute Risk Difference (ARD)

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|             | Infection (+) | Infection (–) | Total |
|-------------|---------------|---------------|-------|
| Exposed     | 100           | 400           | 500   |
| Not Exposed | 30            | 470           | 500   |

### Calculate Absolute Risk Difference (ARD)

- $ARD = \text{Risk in the exposed} - \text{Risk in the unexposed}$

## Interpretation: Absolute Risk Difference (ARD)

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- **ARD =  $0.20 - 0.06 = 0.14$  = 14 percentage points**
- There were **14 more infections per 100 children** in the exposed group than in the unexposed group during follow-up.
- The risk difference describes the **excess burden of disease** associated with passive smoke exposure.

## Poll: Interpreting Relative Risk

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In a cohort study examining **smoking and cardiovascular disease**, the Relative Risk (RR) for smokers compared to non-smokers is **1.50**. What does this mean?

- A.** Smokers have a 50% lower risk of CVD
- B.** Smokers have a 50% higher risk of CVD
- C.** Smokers and non-smokers have the same risk
- D.** Smoking prevents cardiovascular disease

## Poll: Interpreting Risk Difference

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In a cohort study of **smoking and cardiovascular disease (CVD)**, the risks over the same follow-up period are:

- Risk in smokers = **20%**; Risk in non-smokers = **10%**
- Absolute Risk Difference (ARD) = **20% – 10% = 10 percentage points**

What does this ARD mean?

- A.** There were 10 more CVD cases per 100 smokers than per 100 non-smokers during follow-up
- B.** Smoking increases CVD risk by 10 times compared to non-smokers during follow-up
- C.** There was no association between smoking and CVD during follow-up
- D.** 10 out of every 100 cases of CVD were caused by smoking during follow-up

# Cohort Studies: Strengths and Limitations

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## Strengths

- Directly measure **incidence**
- Establish clear **temporal sequence** (Exposure → Outcome)
- Study **rare exposures** efficiently
- Examine **multiple outcomes** from one exposure

## Limitations

- **Time-consuming** and costly
- Vulnerable to **loss to follow-up**
- Potential **exposure misclassification**
- Less efficient for **rare diseases**